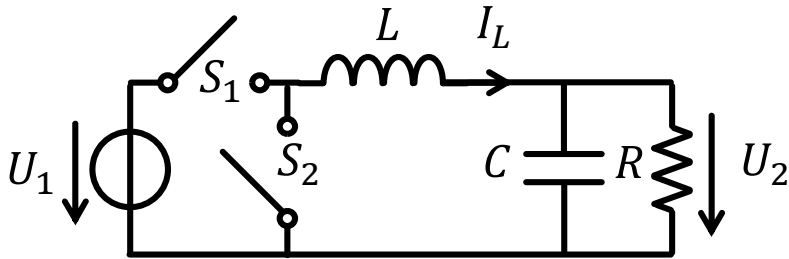


Convertisseur DC-DC BUCK



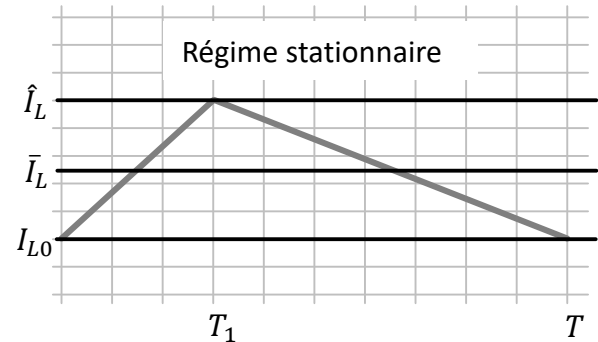
Avec C grande.

$$\Delta U_2 \ll \bar{U}_2$$

$$U_2 \approx \bar{U}_2$$

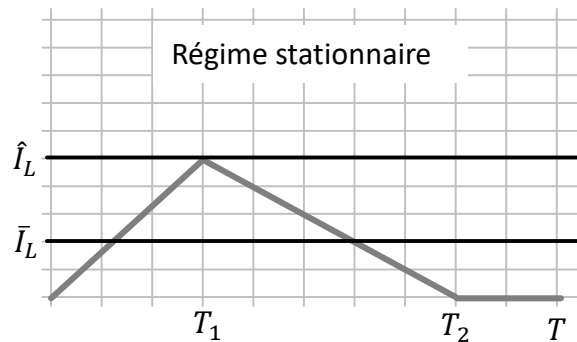
Équations conduction continue

$U_2 = U_1 \times \frac{T_1}{T}$	$\Delta I_L = \frac{(U_1 - U_2)U_2 T}{U_1 L}$
$\bar{I}_L = \bar{I}_R = \frac{U_2}{R}$	$I_{L0} = \bar{I}_L - \frac{\Delta I_L}{2}$
$\Delta U_2 = \Delta V_C = \frac{U_2(U_1 - U_2)}{U_1} \frac{T^2}{8LC}$	$\hat{I}_L = \bar{I}_L + \frac{\Delta I_L}{2}$



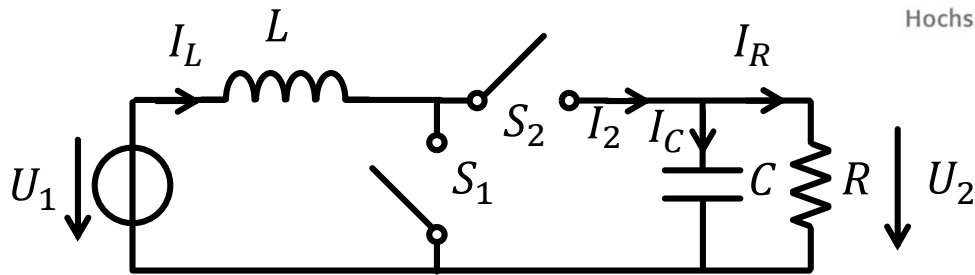
Limite conduction continue / discontinue: $R = \frac{2L}{T} \frac{U_1}{(U_1 - U_2)}$

Équations conduction discontinue



$T_2 = \frac{T_1}{2} \pm \sqrt{\frac{T_1^2}{4} + \frac{2TL}{R}}$	$T_1 = \sqrt{\frac{2TL}{R} \frac{U_2^2}{U_1(U_1 - U_2)}}$
$U_2 = U_1 \frac{T_1}{T_2}$	$T_2 = \frac{U_1}{U_2} T_1$
$\bar{I}_L = \frac{\hat{I}_L T_2}{2T}$	$\bar{I}_L = \frac{U_2}{R}$
$\Delta I_L = \hat{I}_L = \frac{U_1 - U_2}{L} T_1$	$\Delta I_L = \hat{I}_L = \frac{U_2}{L} (T_2 - T_1)$
$\Delta V_C = \Delta U_2 = \frac{(\hat{I}_L - \bar{I}_L)^2 L}{2U_2 C} + \frac{\bar{I}_L^2 L}{2C(U_1 - U_2)} + \frac{(U_1 - U_2) \times T_1^2}{2LC} - \frac{\bar{I}_L}{C} T_1$	

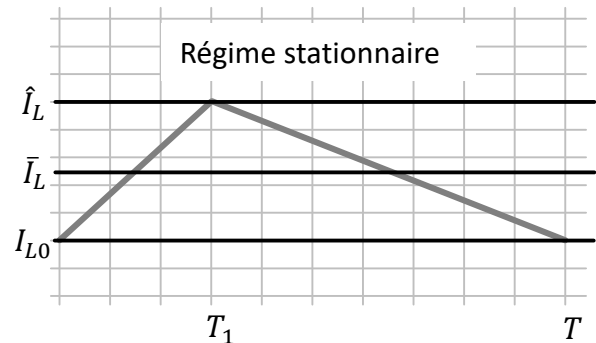
Convertisseur DC-DC BOOST



Avec C grande.
 $\Delta U_2 \ll \bar{U}_2$
 $U_2 \approx \bar{U}_2$

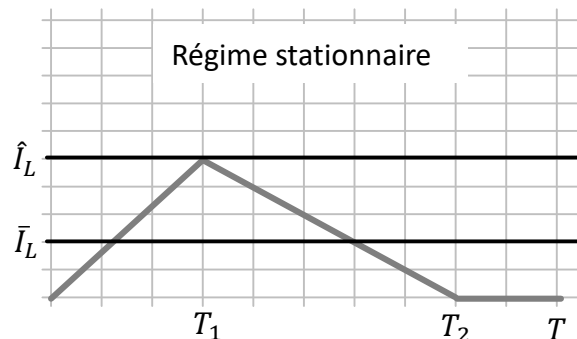
Équations conduction continue

$U_2 = U_1 \frac{T}{T - T_1}$	$T_1 = T \frac{U_2 - U_1}{U_2}$
$\Delta I_L = \frac{U_1(U_2 - U_1) T}{U_2 L}$	$\bar{I}_L = \frac{U_2^2}{R U_1}$
$I_{L0} = \bar{I}_L - \frac{\Delta I_L}{2}$	$\hat{I}_L = \bar{I}_L + \frac{\Delta I_L}{2}$
$\Delta U_2 = \Delta V_C = \frac{(\hat{I}_L - \bar{I}_2)^2 T U_1}{2 \Delta I_L C U_2}$	$\bar{I}_2 = \frac{U_2}{R} = \bar{I}_L \frac{T - T_1}{T}$



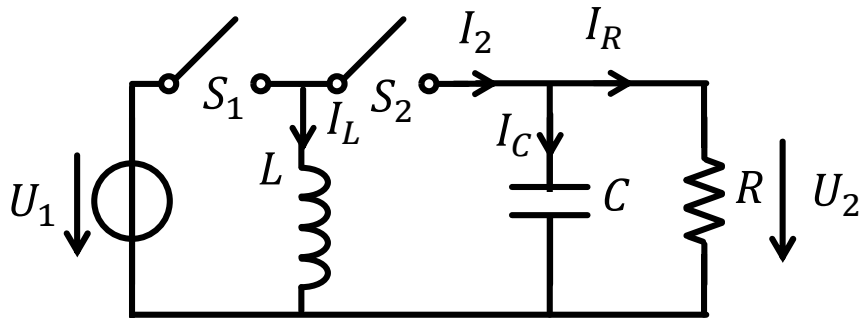
Limite conduction continue / discontinue: $R = \frac{2L}{T} \frac{U_2^3}{U_1^2(U_2 - U_1)}$

Équations conduction discontinue



$U_2 = U_1 \frac{T_2}{T_2 - T_1}$	$T_1 = \sqrt{\frac{2LT}{R} \frac{U_2(U_2 - U_1)}{U_1^2}}$
$\bar{I}_2 = \frac{U_2}{R}$	$T_2 = \frac{U_2}{U_2 - U_1} T_1$
$\bar{I}_L = \frac{\hat{I}_L T_2}{2T}$	$\bar{I}_L = \frac{U_2^2}{U_1 R}$
$\Delta I_L = \hat{I}_L = \frac{U_1}{L} T_1$	$\Delta I_L = \hat{I}_L = \frac{U_2 - U_1}{L} (T_2 - T_1)$
$\Delta U_2 = \Delta V_C = \frac{(\hat{I}_L - \bar{I}_2)^2 T U_1}{2 \Delta I_L C U_2}$	

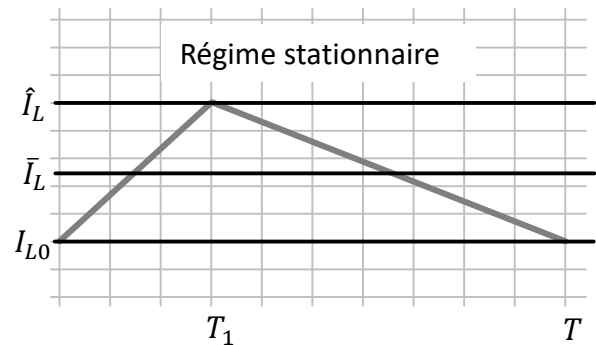
Convertisseur DC-DC BUCK-BOOST



Avec C grande.
 $\Delta U_2 \ll \bar{U}_2$
 $U_2 \approx \bar{U}_2$

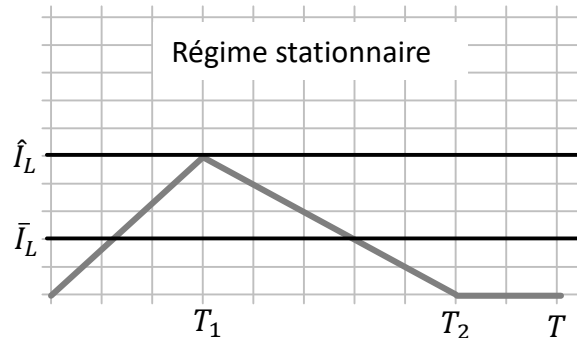
Équations conduction continue

$U_2 = -U_1 \frac{T_1}{T - T_1}$	$T_1 = T \frac{U_2}{U_2 - U_1}$
$\Delta I_L = \frac{U_1 T_1}{L} = -\frac{U_2 (T - T_1)}{L}$	$\bar{I}_L = -\frac{U_2}{R} \frac{T}{T - T_1}$
$I_{L0} = \bar{I}_L - \frac{\Delta I_L}{2}$	$\hat{I}_L = \bar{I}_L + \frac{\Delta I_L}{2}$
$\Delta U_2 = \frac{(\hat{I}_L - \bar{I}_2)^2}{2 \Delta I_L C} \frac{U_1}{U_1 - U_2} T$	$\bar{I}_2 = \frac{U_2}{R} = -\bar{I}_L \frac{T - T_1}{T}$



Limite conduction continue / discontinue: $R = \frac{2L}{T} \frac{(U_2 - U_1)^2}{U_1^2}$

Équations conduction discontinue



$U_2 = -U_1 \frac{T_1}{T_2 - T_1}$	$T_1 = \frac{U_2}{U_1} \sqrt{\frac{2LT}{R}}$
$\bar{I}_2 = \frac{U_2}{R} = -\bar{I}_L \frac{T_2 - T_1}{T_2}$	$T_2 = T_1 \frac{U_2 - U_1}{U_2}$
$\bar{I}_L = \frac{\hat{I}_L T_2}{2T}$	$\bar{I}_L = \frac{U_2 (U_2 - U_1)}{U_1 R}$
$\Delta I_L = \hat{I}_L = \frac{U_1}{L} T_1$	$\Delta I_L = \hat{I}_L = -\frac{U_2}{L} (T_2 - T_1)$
$\Delta U_2 = \Delta V_C = \frac{(\hat{I}_L - \bar{I}_2)^2}{2 \hat{I}_L C} (T_2 - T_1)$	